

HOMEWORK 4 (FP)

PROBLEM I – BUSINESS CYCLES AND NOMINAL RIGIDITIES

Consider a monetary economy with the following markets being open in each period t : a good market with price P_t , a labor market with the wage W_t and a money market. Money is the num'raire (price equals one). The economy is populated by two representative agents that behave in a competitive way: a firm and a household.

The firm has a Cobb-Douglas technology:

$$Y_t = Z_t K_t^\gamma N_t^{1-\gamma} \quad (1)$$

where K_t is capital, N_t labor input, and Z_t a stochastic technological shock. It is assumed that the firm profit Π_t is distributed to the household.

Capital fully depreciates in one period so that

$$K_{t+1} = I_t \quad (2)$$

where I_t is investment in period t .

The representative household works N_t , consumes C_t in period t , and ends the period with a quantity of money M_t . He has the following preferences:

$$U = E_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_t + \omega \log \frac{M_t}{P_t} - V(N_t) \right] \quad (3)$$

where V is a convex function.

At the beginning of period t there is an aggregate stochastic multiplicative monetary shock as in Lucas (1972), denoted by μ_t . The money holdings M_{t-1} carried from the previous period are multiplied by μ_t , so that the household starts period t with money holdings $\mu_t M_{t-1}$.

The household budget constraint in period t is :

$$C_t + \frac{M_t}{P_t} + I_t = \frac{W_t}{P_t} N_t + \kappa_t I_{t-1} + \frac{\mu_t M_{t-1}}{P_t} + \Pi_t \quad (4)$$

where κ_t is the real return in period t on capital invested in $t-1$.

The Walrasian regime (flexible price and wage) :

- 1 – Write down the household maximization program and derive its First Order Conditions.
- 2 – Write down the firm maximization program and derive its First Order Conditions.
- 3 – Define a competitive equilibrium of this economy
- 4 – Show that in equilibrium,

$$C_t = (1 - \beta\gamma)Y_t \quad (5)$$

$$I_t = K_{t+1} = \beta\gamma Y_t \quad (6)$$

$$\frac{M_t}{P_t} = \frac{\omega(1 - \beta\gamma)}{1 - \beta} Y_t = \chi Y_t \quad (7)$$

- 5 – Show that employment is constant along the Walrasian equilibrium path. Denote N this level and n its log.
- 6 – What is the effect of a shock μ_t on output? On prices? Discuss.
- 7 – Show that the walrasian nominal wage w^w and the price level are given by:

$$w_t^w = m_t + \log(1 - \gamma) - \log \chi - n \quad (8)$$

$$p_t = m_t - \frac{z_t}{1 - \gamma L} - \log \chi - n - \frac{\gamma \log \beta\gamma}{1 - \gamma} \quad (9)$$

where L is the lag operator.

- 8 – Compute the correlation between the log of output and the log of the real wage. Discuss.

Nominal Wage Contracts : We now assume that the level of wages is predetermined at the beginning of each period. At this contract wage the household supplies all labor demanded by the firm. The crucial assumption is that parties to the contract aim at clearing the market *ex ante* (in logarithmic terms). In other terms, the contract wage will be set equal to the expected value of the Walrasian wage w^w .

9 – Compute the level of the contract wage as a function of expected money supply, conditionally on $t-1$ information.

Since the goods market clears and the firm's demand for labor is always satisfied, first order conditions of the firm are not affected.

10 – Explain why household now maximize their utility function (3) subject to the budget constraint (4), but taking N_t as *given* (and determined by firms's demands).

11 – Show that equilibrium allocations are now given by

$$y_t = z_t + \gamma k_t + (1 - \gamma)n_t \quad (10)$$

$$w_t - p_t = y_t - n_t + \log(1 - \gamma) \quad (11)$$

$$m_t = p_t + y_t + \log \chi \quad (12)$$

$$k_{t+1} = y_t + \log \beta \gamma \quad (13)$$

plus the equation setting the nominal wage, as computed in question 9.

12 – Show that

$$n_t = n + m_t - E_{t-1}m_t \quad (14)$$

Comment.

Let's define the monetary shock as

$$\varepsilon_{mt} = m_t - E_{t-1}m_t \quad (15)$$

13 – Using the expression of output and real wage (both in logs), show that supply shocks and *lagged* money shocks induce a positive correlation between real wage and output, while *contemporary* money shocks induce inversely a negative correlation between real wage and output. Discuss.

PROBLEM II – RATIONAL EXPECTATIONS AND STAGGERED PRICES

We consider here a model inspired from the works of Stan Fisher (1977) and John Taylor (1978). Consider an economy with a continuum of individuals indexed by i , and uniformly distributed over the interval $[0, 1]$. The advantage of considering a measure 1 of individuals is that, if x_i denotes a individual variable, the aggregate counterpart will be $x = \int_0^1 x_i di$. At a symmetric allocation, we will have $x_i = x \quad \forall i$.

Price setting with imperfect competition and flexible prices: Each individual is the only producer of good i , that is produced in quantity Q_i according to the technology $Q_i = \tilde{L}_i$, where $\tilde{L}_i$ is the amount of labor hired by agent i . Agent i supplies L_i units of labor on a single labor market, and may work in any firm. W is the nominal wage in the economy. Utility of agent i is given by

$$U_i = C_i - \frac{L_i^\gamma}{\gamma}, \quad \gamma > 1. \quad (16)$$

where C_i is agent i consumption. It is a basket of all the goods produced in the economy, with a price $P = \int_0^1 P_i di$. Agent i nominal income I_i is the sum profits $P_i Q_i - W \tilde{L}_i$ and labor income $W L_i$. It is assumed that good i demand is given by

$$Q_i = Y \left(\frac{P_i}{P} \right)^{-\eta}, \quad \eta > 1 \quad (17)$$

where $Y = \int_0^1 Q_i di$ is aggregate production and also aggregate real income. Finally, aggregate demand is given by

$$PY = M \quad (18)$$

where M is the exogenous money supply. Money is the num'raire.

1 – Discuss, interpret, give foundations to equation (18).

2 – Show that the utility maximization problem of agent i reduces to

$$\max_{P_i, L_i} \frac{(P_i - W)Y(P_i/P)^{-\eta} + WL_i}{P} - \frac{L_i^\gamma}{\gamma}$$

3 – Derive the first order conditions of the utility maximization problem. Manipulate those equations to obtain (i) an equation that expresses the relative price P_i/P as a markup over marginal cost and (ii) a labor supply equation that expresses L_i as a function of the real wage.

4 – Solve for the symmetric equilibrium. [Hint: in equilibrium, $x_i = x = \int_0^1 x_i di$ for any variable x of the model, and the good market equilibrium conditions writes $C = Q = Y$.]. Give the equilibrium value of Y and P . Compute the money multiplier and discuss. Discuss of the effect on η on output level.

5 – Define p_i^* as the log of the optimal (flexible) price for agent i . Show that

$$p_i^* = c + (1 - \phi)p + \phi m$$

where $p = \log(P)$, $m = \log(M)$. Give the value of c and ϕ . This price will be referred to as the target price when prices will be predetermined or fixed.

The model with predetermined prices and rational expectations (the Fisher Model): We now assume that half of the agents sets their price in odd periods, half in even ones. When an agent sets prices in period, t , she does set the next period ($t + 1$) price and the price of the period after ($t + 2$), at the expected target levels of period $t + 1$ and $t + 2$. Prices needs not to be the same in periods $t + 1$ and $t + 2$, but they are predetermined. In any period, half of prices are ones set in the previous period and half are ones set two periods ago. Thus, the average (log) price is

$$p_t = \frac{1}{2}(p_t^1 + p_t^2)$$

where p_t^1 denotes the price set for t by individuals who set their prices in $t - 1$ and p_t^2 the price set for t by individuals setting prices in $t - 2$. p_t^1 equals the expectation as of period $t - 1$ of p_{it}^* ($p_t^1 = E_{t-1} p_{it}^*$) and p_t^2 equals the expectation as of $t - 2$ of p_{it}^* ($p_t^2 = E_{t-2} p_{it}^*$). In the following, we consider for simplicity the model without the constant c :

$$p_{it}^* = (1 - \phi)p_t + \phi m_t.$$

Note also that we have added time subscripts.

6 – Express p_t^1 as a function of $E_{t-1} m_t$, p_t^1 and p_t^2 . Express p_t^2 as a function of $E_{t-2} m_t$, $E_{t-2} p_t^1$ and p_t^2 .

7 – Solve for p_t^1 and p_t^2 as a function of $E_{t-1} m_t$ and $E_{t-2} m_t$.

8 – Show that the model solution is given by

$$\begin{cases} p_t &= E_{t-2} m_t + \frac{\phi}{1+\phi} (E_{t-1} m_t - E_{t-2} m_t) \\ y_t &= \frac{1}{1+\phi} (E_{t-1} m_t - E_{t-2} m_t) + (m_t - E_{t-1} m_t) \end{cases}$$

Discuss the economic properties of the solution.

The model with fixed prices and rational expectations (the Taylor model): Assume now that prices are not only predetermined, but fixed for two consecutive periods. To get an easier solution, we slightly change the timing. In period t , half the agents sets the same price χ_t for periods t and $t + 1$, at the expected average target level:

$$\chi_t = \frac{1}{2}(p_{it}^* + E_t p_{it+1}^*)$$

while the other half sets their price in period $t + 1$ for periods $t + 1$ and $t + 2$:

$$\chi_{t+1} = \frac{1}{2}(p_{it+1}^* + E_{t+1} p_{it+2}^*)$$

The average price p_t is therefore given by

$$p_t = \frac{1}{2}(\chi_{t-1} + \chi_t).$$

It is also assumed that m is a random walk:

$$m_t = m_{t-1} + u_t$$

where u is a white noise.

9 – Show that the model equilibrium prices satisfy the recursion

$$\chi_t = A(\chi_{t-1} + E_t \chi_{t+1}) + (1 - 2A)m_t$$

where $A = \frac{1}{2} \frac{1-\phi}{1+\phi}$. Discuss this equation in economic terms.

10 – Assume the solution for χ is of the form $\chi_t = \lambda\chi_{t-1} + (1 - \lambda)m_t$. Use the previous recursion to compute the unique value of λ with modulus smaller than one.

11 – Show that the model solution in output is

$$y_t = \lambda y_{t-1} + \frac{1 + \lambda}{2} u_t$$

Discuss the economic properties of the solution.

PROBLEM III – DISCUSSION – ABOUT GALI'S 1999 AER PAPER (TECHNOLOGY, EMPLOYMENT, AND THE BUSINESS CYCLE: DO TECHNOLOGY SHOCKS EXPLAIN AGGREGATE FLUCTUATIONS?)

In a 1999 AER paper, Jordi Gali is estimating the following VAR:

My empirical model interprets the observed variations in (log) productivity (x_t) and (log) hours (n_t) as originating in two types of exogenous disturbances—technology and non-technology shocks—which are orthogonal to each other, and whose impact is propagated over time through various unspecified mechanisms. That idea is formalized by assuming that the vector $[\Delta x_t, \Delta n_t]'$ can be expressed as a (possibly infinite) distributed lag of both types of disturbances:

$$\begin{aligned} (23) \quad & \begin{bmatrix} \Delta x_t \\ \Delta n_t \end{bmatrix} \\ &= \begin{bmatrix} C^{11}(L) & C^{12}(L) \\ C^{21}(L) & C^{22}(L) \end{bmatrix} \begin{bmatrix} \varepsilon_t^z \\ \varepsilon_t^m \end{bmatrix} \\ &\equiv \mathbf{C}(L) \varepsilon_t \end{aligned}$$

where $\{\varepsilon_t^z\}$ and $\{\varepsilon_t^m\}$ denote, respectively, the sequences of technology and non-technology shocks. The orthogonality assumption (combined with a standard normalization) implies $E\varepsilon_t \varepsilon_t' = I$. Furthermore, the identifying restriction that the unit root in productivity originates exclusively in technology shocks corresponds to $C^{12}(1) = 0$. In other words, the matrix of long-run multipliers $\mathbf{C}(1)$ is assumed to be lower triangular.

The specification in (23) is based on the assumption that both productivity and hours are integrated of order one, so that first-differencing of both variables is necessary to achieve stationarity. That assumption is motivated by the outcome of standard augmented Dickey Fuller (ADF) tests which do not reject the null of a unit root in the levels of either series, but do reject the same null when applied to the first-differences (at the [...])

1 – Explain why it is useful to decompose the VAR innovations into two orthogonal components

2 – Explain what are the assumptions made by Gali to get sequences of technology and non technology shocks. Are these assumptions reasonable?

Some of the results of the estimation are given in the following table. In that table, the correlation between productivity and labor input *conditional on a technology shock* represents the correlation that would be observed if only technology shocks were hitting the economy.

TABLE 1—CORRELATION ESTIMATES: BIVARIATE MODEL

	Unconditional	Conditional	
		Technology	Nontechnology
Panel A: First-differenced labor			
Hours	−0.26** (0.08)	−0.82** (0.12)	0.26** (0.12)
Employment	−0.02 (0.07)	−0.84** (0.26)	0.64** (0.13)
Panel B: Detrended labor			
Hours	−0.26** (0.08)	−0.81** (0.11)	0.35* (0.20)
Employment	−0.02 (0.07)	−0.35 (0.49)	0.38 (0.56)

Notes: Table 1 reports estimates of unconditional and conditional correlations between the growth rates of productivity and labor input (hours or employment) in the United States, using quarterly data for the period 1948:1–1994:4. Standard errors are shown in parentheses. Significance is indicated by one asterisk (10-percent level) or two asterisks (5-percent level). Conditional correlation estimates are computed using the procedure outlined in the text, and on the basis of an estimated bivariate VAR for productivity growth and labor-input growth (Panel A) or productivity growth and detrended labor input (Panel B). Data sources and definitions can be found in the text.

3 – Present in words the results.

4 – What is the effect of a technology shock on labor input (hours or employment) in a plain RBC model? What is the typical shape of a impulse response function to a persistent TFP shock in a plain RBC model?

5 – Consider a fix-price AD-AS model in which Aggregate Demand is given by $Y = \frac{M}{P}$ and the production function by $Y = AH$ where H stands for labor input and A for technology. What would be in such a model the impact of a positive technology shock? What can we conclude from Galí's results?